

A small Game Physics Documentation

This submission is a small rage game, similar to games like only up or getting over it, with the main mechanic of changing your own gravity to get to the goal.

This idea came about while working on another game in the unreal engine (blood and mortar) and rigging the character movements there. By accident, I gave the character some very weird jumps and movements. While they were not wanted in that project, I thought they might be fun as mechanics in a different game. And here we are.

The original goal was to make a regular jump and run with gravity changing and try to find jumps that are small enough to not lose health. However, I quickly noticed that moving your character on surfaces that are not the ground is very counter-intuitive. This made me drop the health mechanic entirely and instead shift more to the rage-game rage game mechanics of unintuitive or difficult movement.

After implementing the gravity change, the first thing I noticed while testing is that you could, in theory, fly infinitely by rapidly switching the direction of gravity. This was obviously not a mechanic that works well for a jump and run (although it also sounds like a fun main mechanic for a different game). So, to prevent this, I changed the gravity to only be changeable on solid ground.

The second thing was that regular movement is really not made to work on walls. This was slightly tricky, because while walking on the wall sideways worked, up and down did not work, since the movement didn't account for the Z-axis. Unfortunately I did not find a better fix to this problem than to bind the player movement to the camera. This makes the movement even more wonky than it was before. On the other hand, it also adds to the rage-game mechanics.

The changing gravity intensity was added because that was the original mess up in blood and mortar and to make for some interesting jumps. While lower gravity makes all jumps easier, it is also intentionally slow, so that overusing it doesn't make for as engaging of a gameplay loop as only using it at the points it is intended to be used.

I wanted to implement some kind of compass that shows you which direction you are looking in, but I knew no easy way to do that.

All jumps are tested and possible, even for someone like me, who usually doesn't play jump and runs.